

Every operator in a singleton image is compact

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.isCompactOperator_map_of_singleton`

theorem

Excludes a noncompact image by making one separating character take both zero and one on the same projection.

Statement

Let A be a nonzero complex C^* -algebra, with no unit assumed, let H be a separable complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ represent the unique unitary-equivalence class of nonzero irreducible $*$ -representations of A . Then

$$\pi(a) \in \mathcal{K}(H) \quad (a \in A).$$

Assumptions

No simplicity or separability of A is assumed. The argument uses faithfulness of π on A , not faithfulness of its unital extension on the whole unitization.

Conclusion

This proves the inclusion $\pi(A) \subseteq \mathcal{K}(H)$ for the specified representation.

Proof route

Separate a noncompact self-adjoint image from the compact-preimage ideal in a maximal abelian subalgebra. Its character produces a rank-one projection which belongs to the ideal but has character value one.

▼ Proof steps

1. Suppose $\pi(a)$ is noncompact. Write

$$a = \operatorname{Re} a + i \operatorname{Im} a, \quad \operatorname{Re} a = (a + a^*)/2, \quad \operatorname{Im} a = (a - a^*)/(2i).$$

If both self-adjoint parts had compact images, their sum would too. Choose a self-adjoint part $b \in A$ with noncompact $\pi(b)$. Write $\tilde{A} = \mathbb{C} \oplus A$, $\iota(x) = (0, x)$, $\chi_\infty(\lambda, x) = \lambda$, and $\hat{\pi}(\lambda, x) = \lambda I + \pi(x)$. Apply A maximal abelian subalgebra containing a self-adjoint element to the self-adjoint $\iota(b)$ in the unital $\tilde{A}$, obtaining a maximal abelian unital $*$ -subalgebra D containing it. Such a D is norm closed.

2. For the representation $\rho_D = \hat{\pi}|_D$, form The compact-preimage ideal and use Norm closedness of the compact-preimage ideal to obtain the closed ideal

$$J = \{d \in D : \widehat{\pi}(d) \text{ is compact}\}.$$

Commutativity makes it an ordinary ideal of the commutative C^* -algebra D . The element $\iota(b) \in D$ is outside J , since $\widehat{\pi}(\iota(b)) = \pi(b)$. Apply A character separating a closed ideal to J , $\iota(b)$: obtain one character χ with $\chi|_J = 0$ and $\chi(\iota(b)) \neq 0$.

3. The scalar character vanishes at $\iota(b)$, so $\chi \neq \chi_\infty|_D$. The closedness of D and the singleton assumptions allow A unit eigenvector for a non-scalar character to give a unit vector $\eta \in H$ with $\widehat{\pi}(d)\eta = \chi(d)\eta$ for all $d \in D$. Apply Preimages of all rank-one operators to η, η , using separability of H , and obtain $p \in A$ with

$$\pi(p) = \theta_{\eta, \eta}, \quad \theta_{\eta, \eta}(z) = \langle \eta, z \rangle \eta.$$

4. For $d \in D$, both d and d^* lie in D , so

$$\widehat{\pi}(d)\eta = \chi(d)\eta, \quad \widehat{\pi}(d)^*\eta = \overline{\chi(d)}\eta.$$

Fix any $z \in H$ and use the inner product linear in its second argument. The two products are

$$\widehat{\pi}(d)\theta_{\eta, \eta}z = \langle \eta, z \rangle \widehat{\pi}(d)\eta = \chi(d)\langle \eta, z \rangle \eta,$$

and

$$\begin{aligned} \theta_{\eta, \eta}\widehat{\pi}(d)z &= \langle \eta, \widehat{\pi}(d)z \rangle \eta \\ &= \langle \widehat{\pi}(d)^*\eta, z \rangle \eta \\ &= \overline{\chi(d)}\langle \eta, z \rangle \eta. \end{aligned}$$

The last equality uses conjugate linearity in the first argument. Thus these operators commute.

5. Write the same d as (λ, x) with $x \in A$. Substituting $\widehat{\pi}(d) = \lambda I + \pi(x)$ and $\theta_{\eta, \eta} = \pi(p)$ in the commutation identity and cancelling the scalar terms gives

$$\pi(xp) = \pi(x)\pi(p) = \pi(p)\pi(x) = \pi(px).$$

By Faithfulness on the original algebra, the original π is injective; hence $xp = px$. Consequently

$$d\iota(p) = (0, \lambda p + xp) = (0, \lambda p + px) = \iota(p)d.$$

This holds for every $d \in D$, so The commutant of a maximal abelian subalgebra gives $\iota(p) \in D$. No injectivity of $\widehat{\pi}$ is used.

6. The represented operator of $\iota(p) \in D$ is $\pi(p) = \theta_{\eta, \eta}$, which is compact. Thus $\iota(p) \in J$ and

$$\chi(\iota(p)) = 0.$$

On the other hand, the eigenvector equation for this very element gives

$$\begin{aligned} \chi(\iota(p))\eta &= \widehat{\pi}(\iota(p))\eta \\ &= \theta_{\eta, \eta}\eta = \langle \eta, \eta \rangle \eta = \eta. \end{aligned}$$

Since $\|\eta\| = 1$, this implies $\chi(\iota(p)) = 1$, contradicting its value zero. The original noncompact image is therefore impossible.

Main citations

- Exact theorem and proof
- A maximal abelian subalgebra containing a self-adjoint element
- The compact-preimage ideal
- Norm closedness of the compact-preimage ideal
- A character separating a closed ideal
- A unit eigenvector for a non-scalar character
- Preimages of all rank-one operators
- Faithfulness on the original algebra
- The commutant of a maximal abelian subalgebra

Lean source signature (exact)

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theorem isCompactOperator_map_of_singleton [Nontrivial A]
  [TopologicalSpace.SeparableSpace H]
  (pi : NonUnitalCStarRepresentation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi) :
  ∀ a : A, IsCompactOperator (pi a)
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In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
[TopologicalSpace.SeparableSpace H]	The complex Hilbert space H is separable; this is used for the rank-one preimage theorem.
(pi : NonUnitalCStarRepresentation A H)	The specified $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$; no unit-preservation equation is required.
(hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.
∀ a : A, IsCompactOperator (pi a)	For every element of the original A , its represented operator on H is compact. This is one inclusion of ranges, not the assertion that every bounded operator is compact.

Exact source and proof.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.isCompactOperator_map_of_singleton

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